

GIAIC Final Graduation Exam Preparation

AI Agent Factory (Official Book)

Module 3 Quiz – Local AI and Agentic Coding

50 Important Exam MCQs with Answers

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Q1. A developer has already configured Claude Code as a coding agent. Instead of changing Claude Code itself, she changes its endpoint from a local Ollama server running on her laptop to a vLLM server running on a GPU machine. The user interface, workflow, and coding-agent program remain unchanged. Which architectural component has primarily changed?
a. The harness **b.** The address **c.** The tool-call schema **d.** The user's IDE

Answer: b. The address

Q2. The Agent Factory material describes an AI coding system using the concepts of a **harness** and a **brain**. Suppose the harness is responsible for reading project files, executing commands, and displaying changes, while the model is responsible for reasoning about what should happen next. Which statement most accurately describes their relationship?
a. The harness is the model itself, while the brain is only the terminal interface
b. The brain executes filesystem operations directly, while the harness only displays results
c. The harness provides the working environment and tools, while the brain supplies the reasoning that determines what to do
d. The harness and brain are interchangeable names for the same software component

Answer: c. The harness provides the working environment and tools, while the brain supplies the reasoning that determines what to do

Q3. A student says, “If I install Ollama, I have automatically created a powerful coding agent.” Another student explains that Ollama primarily provides a way to **serve or run a model**, while a coding agent such as Claude Code or OpenCode provides the surrounding

tool-use workflow. Which explanation is most accurate?

- a. Ollama itself is always a complete autonomous coding agent
- b. Ollama is a serving layer, while a coding agent can act as the harness around the model
- c. Ollama is only a cloud gateway and cannot execute models locally
- d. Claude Code is the serving layer and Ollama is only a text editor

Answer: b. Ollama is a serving layer, while a coding agent can act as the harness around the model

Q4. A developer downloads an open-weight model through Ollama. She then disconnects the computer from Wi-Fi and asks the same model another question. The model responds normally. Which conclusion is most justified?

- a. The request was secretly sent to OpenRouter through a cached browser connection
- b. The local model is capable of running on the machine without requiring an internet connection for inference
- c. Ollama automatically converted the model into a browser-based AI
- d. The model must have been running on a remote GPU cluster

Answer: b. The local model is capable of running on the machine without requiring an internet connection for inference

Q5. A developer has a local coding agent that takes several minutes to generate each response, but when the response finally arrives, the model correctly understands the task and produces valid tool calls. Which problem is more likely being demonstrated?

- a. The capability wall, because the model cannot understand coding tasks
- b. The throughput or hardware wall, because the model is capable but the machine is too slow
- c. The authentication wall, because local models require cloud credentials
- d. The address wall, because localhost cannot be used by coding agents

Answer: b. The throughput or hardware wall, because the model is capable but the machine is too slow

Q6. Another developer has a powerful GPU, and the model responds quickly, but the coding agent repeatedly generates malformed structured tool calls such as invalid file-edit instructions. Why would adding an even faster GPU fail to directly solve the fundamental problem?

- a. GPU performance determines whether a model understands tool-call formats
- b. The issue is primarily model capability and tool-use reliability rather than raw inference throughput
- c. Faster GPUs automatically reduce context windows

d. Tool calls can only be generated through OpenRouter

Answer: b. The issue is primarily model capability and tool-use reliability rather than raw inference throughput

Q7. Consider two local setups. Setup A uses a stronger model on slow hardware and produces correct results after several minutes. Setup B uses a weaker model on powerful hardware and responds almost instantly but frequently makes incorrect edits. Which conclusion best reflects the two-wall concept?

- a. Setup A mainly demonstrates a throughput limitation, while Setup B mainly demonstrates a capability limitation
- b. Setup A demonstrates a capability limitation, while Setup B demonstrates an internet limitation
- c. Both setups demonstrate exactly the same limitation
- d. Neither setup has a meaningful limitation because both are local

Answer: a. Setup A mainly demonstrates a throughput limitation, while Setup B mainly demonstrates a capability limitation

Q8. A developer decides to upgrade from a small model to a more capable model because the existing model repeatedly misunderstands multi-step coding instructions. However, after the upgrade, response times become much slower on the same laptop. What trade-off has most likely occurred?

- a. Model capability improved, but the hardware now struggles to provide acceptable throughput
- b. Hardware capability improved, but model capability decreased
- c. The developer changed from local inference to cloud inference automatically
- d. The harness stopped supporting tool calls

Answer: a. Model capability improved, but the hardware now struggles to provide acceptable throughput

Q9. A company wants an AI coding system for 50 developers. Each developer should be able to submit requests concurrently to the same model, and the organization owns or rents a GPU server. Which architecture best matches this requirement?

- a. Separate Ollama installations on 50 laptops with no shared serving layer

- b. A shared vLLM-based server capable of serving multiple concurrent requests
- c. A single local model with Wi-Fi disabled
- d. A text-generation browser extension with no model server

Answer: b. A shared vLLM-based server capable of serving multiple concurrent requests

Q10. Why does the Agent Factory material intentionally keep a model such as Qwen3 8B constant when comparing the local and server tiers?

- a. To prove that the model becomes more intelligent when moved to a server
- b. To isolate the effect of the serving environment and hardware rather than confusing it with a model change
- c. Because Qwen3 8B cannot run through Ollama
- d. Because vLLM only supports exactly one model

Answer: b. To isolate the effect of the serving environment and hardware rather than confusing it with a model change

Q11. A coding agent sends a request to a model asking it to modify `app.py`. Instead of returning a natural-language sentence such as “I will modify the file,” the model produces a structured instruction containing the file path and requested operation. Why is this distinction important in agentic coding?

- a. Structured tool calls provide machine-readable instructions that the harness can execute
- b. Natural language cannot ever be understood by a coding agent
- c. Tool calls are only used to improve GPU utilization
- d. Structured tool calls automatically make a weak model more intelligent

Answer: a. Structured tool calls provide machine-readable instructions that the harness can execute

Q12. Suppose a local model correctly reasons that a file should be edited and produces a valid `edit_file` tool call. Who is responsible for actually applying the modification to the filesystem in a typical coding-agent architecture?

- a. The model directly changes the bytes on disk
- b. The harness receives and executes the tool call
- c. The GPU writes the source code directly
- d. The API key performs the filesystem operation

Answer: b. The harness receives and executes the tool call

Q13. A model correctly understands that `server.py` needs to be changed but returns a malformed structured tool call that does not match the format expected by the coding agent. The harness therefore cannot execute the requested operation. What has primarily failed?

- a. The model-to-tool interface contract
- b. The GPU memory allocator
- c. The OpenRouter billing system
- d. The Git repository's branch system

Answer: a. The model-to-tool interface contract

Q14. Why does the Agent Factory material recommend performing coding-agent experiments inside a **throwaway Git folder** rather than immediately pointing an experimental local agent at an important production repository?

- a. Throwaway folders make models generate tokens faster
- b. They reduce the potential damage caused by incorrect edits or commands produced by an experimental agent
- c. Git is required for Ollama to download models
- d. Production repositories cannot contain Python files

Answer: b. They reduce the potential damage caused by incorrect edits or commands produced by an experimental agent

Q15. A user gives a coding agent the instruction: “Inspect this project, find a small safe improvement, implement it, and show me what changed.” Which sequence most accurately describes the agentic workflow?

- a. User → model → GPU → GitHub → user
- b. User → harness → model reasoning → structured tool calls → harness executes tools → results returned to model/user
- c. User → OpenRouter → browser → filesystem without a harness
- d. User → GPU → operating system → model → user

Answer: b. User → harness → model reasoning → structured tool calls → harness executes tools → results returned to model/user

Q16. The term `localhost` is frequently misunderstood by beginners. In the context of local AI serving, what does `localhost` most accurately indicate?

- a. A public IP address belonging to the model provider
- b. The same computer on which the request is being made
- c. A special OpenRouter cloud endpoint
- d. A GPU cluster shared by multiple users

Answer: b. The same computer on which the request is being made

Q17. A developer changes an AI coding agent's base URL from a remote provider to `http://localhost:11434`. Assuming Ollama is listening there and the model is available, what is the architectural consequence?

- a. The harness now attempts to reach a model-serving process on the same machine
- b. The harness automatically becomes an Ollama server
- c. The model is uploaded to GitHub
- d. The request is necessarily routed through OpenRouter

Answer: a. The harness now attempts to reach a model-serving process on the same machine

Q18. A student argues, “Open-weight and open-source mean exactly the same thing, so if a model's weights are downloadable, every aspect of its training process and licensing must also be open.” What is the best correction?

- a. Correct; downloadable weights guarantee complete openness
- b. Open-weight primarily means the trained weights can be downloaded; training data, processes, or licensing terms may not all be open in the formal sense
- c. Open-weight means the model cannot be modified
- d. Open-source models can never be downloaded

Answer: b. Open-weight primarily means the trained weights can be downloaded; training data, processes, or licensing terms may not all be open in the formal sense

Q19. A student downloads a large open-weight model and assumes, “Because the weights are freely downloadable, my ordinary laptop must be able to run it comfortably.” Which response best reflects the course's lesson?

- a. Correct, because open-weight models are designed specifically for laptops
- b. Incorrect, because model availability does not guarantee that a particular machine has enough memory or compute to serve it effectively
- c. Correct, because Ollama automatically provides unlimited GPU memory
- d. Incorrect only because open-weight models require an internet connection

Answer: b. Incorrect, because model availability does not guarantee that a particular machine has enough memory or compute to serve it effectively

Q20. The course identifies two separate “walls” that a local coding-agent setup must clear. Which pair correctly represents those walls?

- a.** Privacy and licensing **b.** Model capability and hardware/throughput **c.** Internet speed and storage capacity **d.** API authentication and cloud billing

Answer: b. Model capability and hardware/throughput

Q21. A developer uses a highly capable model that understands complex repository-level instructions, but the model takes so long to process each step that the agent becomes impractical for interactive development. Which statement best explains the problem?

- a.** High capability automatically guarantees high throughput **b.** Capability and throughput are separate concerns; a capable model can still be too slow on inadequate hardware **c.** The model must therefore be closed-source **d.** The harness is necessarily defective

Answer: b. Capability and throughput are separate concerns; a capable model can still be too slow on inadequate hardware

Q22. A model generates responses extremely quickly but cannot reliably decide which file to modify, frequently invents nonexistent APIs, and fails at multi-step coding tasks. Increasing GPU throughput alone would have little value because:

- a.** The dominant limitation is model capability rather than raw generation speed **b.** GPU throughput only matters for cloud models **c.** Faster GPUs reduce the model's reasoning ability **d.** Local models cannot perform multi-step tasks

Answer: a. The dominant limitation is model capability rather than raw generation speed

Q23. Which pairing correctly identifies an appropriate remedy for each wall?

- a.** Capability → use a more capable model; throughput → use better hardware/serving infrastructure **b.** Capability → increase internet speed; throughput → change the model's name **c.** Capability → add more Git branches; throughput → increase API-key length **d.** Capability → use localhost; throughput → use natural language instead of tools

Answer: a. Capability → use a more capable model; throughput → use better hardware/serving infrastructure

Q24. A developer moves the exact same model from Ollama on a laptop to vLLM on a powerful GPU server. The model now answers substantially faster under load, but its underlying learned knowledge and reasoning capability have not magically changed merely because the serving layer changed. What does this demonstrate?

- a.** Serving infrastructure and model capability are conceptually distinct **b.** vLLM retrains every model automatically **c.** Ollama permanently reduces model intelligence **d.** GPU servers convert open models into proprietary models

Answer: a. Serving infrastructure and model capability are conceptually distinct

Q25. A team compares the same model under two environments. With Ollama, 50 requests arrive and many spend time waiting. With vLLM, the same requests are processed much more efficiently. What is the most reasonable interpretation?

- a.** The model itself became smarter under vLLM **b.** The serving layer is better suited to high-concurrency workloads **c.** Ollama cannot perform inference **d.** vLLM automatically increases the model's parameter count

Answer: b. The serving layer is better suited to high-concurrency workloads

Q26. In the context of this module, what does **concurrency** most directly describe?

- a.** The total number of parameters contained in a model **b.** The number of requests being handled or arriving at approximately the same time **c.** The number of tokens contained in a context window **d.** The number of tools installed in the coding agent

Answer: b. The number of requests being handled or arriving at approximately the same time

Q27. A classroom of 50 students simultaneously submits one request each to a shared model server. Ignoring timing nuances, what concurrency level is this workload intended to illustrate?

- a.** 1 **b.** 5 **c.** 50 **d.** 500

Answer: c. 50

Q28. Suppose a server processes requests from many users simultaneously. The organization wants to know how much useful token-generation work the entire system performs per second across all active users. Which metric is most relevant?

- a. Model parameter count b. Aggregate throughput c. Context-window size d. API-key length

Answer: b. Aggregate throughput

Q29. Why is a serving layer designed primarily around a single person's local use case not necessarily the best architecture for a classroom of 50 simultaneous users?

- a. A local serving layer can become a bottleneck and queue requests rather than efficiently exploiting a shared GPU for many concurrent requests b. Local models cannot generate more than one token c. Local AI always requires an internet connection d. A single-user serving layer automatically deletes additional requests

Answer: a. A local serving layer can become a bottleneck and queue requests rather than efficiently exploiting a shared GPU for many concurrent requests

Q30. A traditional batch-processing approach waits for a fixed group of requests to finish together before moving on. vLLM's **continuous batching** instead allows new requests to enter processing as capacity becomes available. What is the primary advantage?

- a. It keeps the GPU more continuously utilized while serving changing workloads b. It makes every model logically more intelligent c. It removes the need for model weights d. It converts every request into an API key

Answer: a. It keeps the GPU more continuously utilized while serving changing workloads

Q31. Imagine three requests: A is almost finished, B has just started, and C arrives while A is still generating. Under an effective continuous-batching strategy, what is the key idea?

- a. C must always wait until every earlier request has completely finished before being considered b. The serving system can dynamically use available processing capacity for C while continuing to serve active requests c. C is automatically rejected because concurrency cannot change d. C is sent to Ollama regardless of the configured server

Answer: b. The serving system can dynamically use available processing capacity for C while continuing to serve active requests

Q32. Why is efficient GPU-memory management especially important when many conversations are active simultaneously?

- a.** Every conversation may require working memory for its active model state, so inefficient allocation can waste scarce GPU memory
- b.** GPU memory is only used to store the user's username
- c.** Context has no relationship to memory consumption
- d.** More conversations always reduce memory usage

Answer: a. Every conversation may require working memory for its active model state, so inefficient allocation can waste scarce GPU memory

Q33. PagedAttention is discussed in the broader vLLM serving architecture because efficient memory allocation matters when serving many concurrent requests. Which description best captures its relevance?

- a.** It is primarily a method for managing attention-related memory more efficiently rather than simply reserving one giant contiguous memory block per request
- b.** It is an API authentication mechanism
- c.** It converts local models into cloud models
- d.** It increases the number of model parameters

Answer: a. It is primarily a method for managing attention-related memory more efficiently rather than simply reserving one giant contiguous memory block per request

Q34. Which comparison most accurately captures the intended distinction between Ollama and vLLM in this module?

- a.** Ollama is primarily for cloud routing, while vLLM is primarily for desktop chat
- b.** Ollama is convenient for running models locally for an individual, while vLLM is designed to serve models efficiently to many users on serious GPU infrastructure
- c.** Both are coding-agent harnesses with identical responsibilities
- d.** Both are cloud-only model marketplaces

Answer: b. Ollama is convenient for running models locally for an individual, while vLLM is designed to serve models efficiently to many users on serious GPU infrastructure

Q35. A school wants to provide one centrally managed AI coding model to 50 students. Each student should connect to the same infrastructure rather than installing a large model independently. Which architecture is most appropriate?

- a.** One separate local Ollama instance per student, with no central server
- b.** A shared GPU server/cluster with an appropriate serving layer such as vLLM
- c.** One browser tab

containing a static answer sheet **d.** A local model on the teacher's laptop with no network access

Answer: b. A shared GPU server/cluster with an appropriate serving layer such as vLLM

Q36. A company wants to run many coding agents continuously and has access to a sufficiently powerful GPU server. Why can a shared server architecture be preferable to forcing every developer to own powerful hardware?

a. One capable server can be shared among many users, potentially improving hardware utilization and centralizing infrastructure **b.** Server GPUs eliminate all AI costs **c.** A server automatically makes every model open-source **d.** Shared servers never experience concurrency problems

Answer: a. One capable server can be shared among many users, potentially improving hardware utilization and centralizing infrastructure

Q37. What is the most accurate description of **OpenRouter** in the architecture presented in Module 3?

a. A local operating system for running Ollama **b.** A cloud gateway that provides access to many hosted models through a common interface **c.** A replacement for Git **d.** A GPU memory-management library

Answer: b. A cloud gateway that provides access to many hosted models through a common interface

Q38. A student says, “Because OpenRouter gives me access to a model, OpenRouter must be the physical GPU cluster that runs every model.” Why is this statement inaccurate?

a. OpenRouter functions as a gateway, while underlying providers can operate the actual model-serving infrastructure **b.** OpenRouter does not communicate with models at all **c.** OpenRouter only works with locally installed models **d.** OpenRouter is itself a GPU hardware manufacturer

Answer: a. OpenRouter functions as a gateway, while underlying providers can operate the actual model-serving infrastructure

Q39. A developer wants to experiment with a very large frontier open-weight model but does not own a GPU cluster capable of serving it. Which tier most directly solves the infrastructure

problem?

- a. Local Ollama on a basic laptop b. A cloud gateway such as OpenRouter that provides access to hosted models c. A larger Git repository d. A longer system prompt

Answer: b. A cloud gateway such as OpenRouter that provides access to hosted models

Q40. A developer has one coding-agent harness and wants to use a small local model for inexpensive routine tasks but a much more capable cloud model for difficult reasoning tasks. Which architectural idea makes this possible?

- a. The harness must be completely rewritten every time the model changes b. The harness can be pointed toward different model addresses according to the task c. Local models automatically become cloud models during difficult tasks d. The model must always be physically installed on the same computer as the harness

Answer: b. The harness can be pointed toward different model addresses according to the task

Q41. Which sequence correctly represents the three serving tiers emphasized by the course?

- a. Local with Ollama → Server/Cluster with vLLM → Cloud through OpenRouter b. Cloud with OpenRouter → Local with vLLM → Server with Ollama c. Local with OpenRouter → Cloud with Ollama → Server with Git d. Server with Claude Code → Local with OpenRouter → Cloud with vLLM

Answer: a. Local with Ollama → Server/Cluster with vLLM → Cloud through OpenRouter

Q42. A coding agent is configured with a particular model but the developer changes only the endpoint from a laptop-hosted model to a server-hosted model. The agent still performs the same high-level workflow of reading files, making plans, issuing tool calls, and displaying results. Which principle does this illustrate?

- a. The harness can remain stable while the address used to reach the brain changes b. Every model must have its own unique coding-agent program c. Server models cannot be used with coding agents d. The filesystem itself becomes the AI brain

Answer: a. The harness can remain stable while the address used to reach the brain changes

Q43. The phrase “**swappable brain**” is used to explain an important architectural property of agentic systems. Which interpretation is the strongest?

- a.** The reasoning model can potentially be replaced while preserving much of the surrounding harness and workflow, provided the interface is compatible **b.** The physical GPU is removed and replaced every time a prompt is submitted **c.** The coding agent itself develops a new neural network after every request **d.** All models contain identical learned parameters

Answer: a. The reasoning model can potentially be replaced while preserving much of the surrounding harness and workflow, provided the interface is compatible

Q44. A company can access a very large model through a cloud gateway, but its own server has insufficient GPU memory to host that model comfortably. Which statement best describes the advantage of the cloud tier?

- a.** Cloud access allows the company to use hosted infrastructure without having to purchase enough local hardware to self-host that model **b.** Cloud access makes the model's weights smaller **c.** Cloud access eliminates the model's computational requirements **d.** Cloud access converts the model into a local Ollama process

Answer: a. Cloud access allows the company to use hosted infrastructure without having to purchase enough local hardware to self-host that model

Q45. The course distinguishes between durable architectural concepts and rapidly changing implementation details. Which item is most likely a **mechanical detail** that should be verified from current documentation rather than treated as a permanent architectural principle?

- a.** The general idea that a model can be reached through an address **b.** The distinction between a harness and a brain **c.** An exact command flag, model tag, software version, or current pricing value **d.** The concept of Local versus Server versus Cloud deployment

Answer: c. An exact command flag, model tag, software version, or current pricing value

Q46. A developer exposes a self-hosted vLLM server to multiple users. She then places an API key in a public Git repository so that clients can connect easily. What is the most serious problem with this approach?

- a.** API keys are harmless because they only identify the model **b.** The credential can be exposed and potentially abused, especially when cloud billing or access control is involved **c.** Git automatically converts API keys into model weights **d.** vLLM requires all API

keys to be publicly visible

Answer: b. The credential can be exposed and potentially abused, especially when cloud billing or access control is involved

Q47. A developer is deciding between the three tiers. Her primary requirement is that sensitive project information remain on her own machine, she has modest workload requirements, and she accepts slower performance. Which choice best aligns with those priorities?

- a.** Local AI with Ollama **b.** A shared public cloud gateway **c.** A frontier model through OpenRouter **d.** A large multi-user vLLM cluster

Answer: a. Local AI with Ollama

Q48. Another organization has hundreds of requests arriving concurrently, owns suitable GPU infrastructure, and wants to maximize useful processing across those requests rather than simply optimize for a single user's interactive experience. Which choice is most appropriate?

- a.** Local Ollama on one employee's laptop **b.** vLLM on a suitable GPU server or cluster
c. Offline text generation with no serving layer **d.** A local browser cache

Answer: b. vLLM on a suitable GPU server or cluster

Q49. A developer needs access to a frontier open model that is far too large for her laptop and does not want to purchase or maintain the necessary GPU cluster. She is willing to pay per usage and wants to connect her existing coding agent through a common gateway. Which architecture best satisfies the requirements?

- a.** Local Ollama **b.** Self-hosted vLLM on the laptop **c.** OpenRouter cloud access
d. Offline CPU inference

Answer: c. OpenRouter cloud access

Q50. A team concludes: “Our local coding agent is slow, so we should simply replace the model with a more intelligent one. If it is still slow, we should keep replacing models until performance improves.” Based on the complete Module 3 architecture, which response is the most technically accurate?

- a.** Correct, because model intelligence is the only factor determining agent performance
b. Incorrect, because model capability and serving throughput are separate concerns; a

stronger model may improve capability, while stronger hardware or a more suitable serving layer may be required to improve throughput **c.** Correct, because vLLM and Ollama change the model's intelligence rather than its serving behavior **d.** Incorrect, because local AI cannot be used for coding agents at all

Answer: b. Incorrect, because model capability and serving throughput are separate concerns; a stronger model may improve capability, while stronger hardware or a more suitable serving layer may be required to improve throughput

Thank you!